Personal details and date of CV

- **Surname:** Liu
- **First names:** Zhuchenyang
- **Researcher ID (ORCID):** 0009-0003-5034-2551
- **Links:** LinkedIn Profile

- **Date of CV:** March 21, 2026

Degree

- **2025-06-18: Master of Science in Data Science (Dual Degree)**
  - KTH Royal Institute of Technology (Sweden) & Aalto University (Finland) (EIT Digital Master School).
  - *Major:* Data Science. *GPA:* 4.56/5.0.
  - Achieved maximum grades in all technical courses.
- **2023-06: Bachelor of Science**
  - Jiangxi University of Finance and Economics (China).
  - *Major:* Data Science and Big Data Technology.

Current employment

- **2025-10 – Present: Doctoral Researcher (Fixed-term)**
  - **Employer:** Aalto University, Department of Information and Communications Engineering, Espoo, Finland.
  - **Supervisor:** Prof. Yu Xiao.
  - **Research Topic:** Reliable Vision-based Complex Document Understanding via Multimodal Large Language Models.
  - **Key Project (AiWo):** Lead Researcher for NLP and Instruction Generation, collaborating with Finnish industrial partners. Designed a decoupled Multimodal Retrieval-Augmented Generation (Visual RAG) framework using ColPali. Successfully implemented an end-to-end local prototype for indexing and inference.
  - **Stage of academic research career:** Stage I (Doctoral Researcher).

Previous work experience

- **2025-02 – 2025-08: Master Thesis Intern**
  - **Employer:** ABB Corporate Research, Västerås, Sweden.
  - *Description:* Developed "AcademicRAG", a Graph-based RAG framework. Proposed a novel clues-based retrieval algorithm utilizing Knowledge Graphs and LLMs, outperforming SOTA baselines in academic Q&A benchmarks.
- **2022-06 – 2022-08: System Architecture Intern**
  - **Employer:** Whale Cloud Technology Co., Ltd, China.
  - *Description:* Assisted in designing high-level architecture for multimodal search functions in a smart city intelligent system.

Research output

Publications

- **1. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *IKEA-Bench: Benchmarking and Mechanistic Analysis of Vision-Language Models for Cross-Depiction Assembly Instruction Alignment*. arXiv preprint arXiv:2604.00913. (Under Review).
  - *Role:* First author. Constructed the benchmark and conducted the three-layer mechanistic analysis of VLMs.

- **2. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *NanoVDR: Distilling a 2B Vision-Language Retriever into a 70M Text-Only Encoder for Visual Document Retrieval*. arXiv preprint arXiv:2603.12824. (Under Review).
  - *Role:* First author. Proposed an asymmetric distillation framework achieving 95% teacher quality with 32× fewer parameters.
- **3. Liu, Z.**, Hu, Z., Zhang, Y. & Xiao, Y. (2026). *Look in the Middle: Structural Anchor Pruning for Scalable Visual RAG Indexing*. arXiv preprint arXiv:2601.20107. (Under Review).
  - *Role:* First author. Conceived the structural anchor pruning method to optimize Visual RAG indexing efficiency.
- **4. Liu, Z.**, Zhang, Y., He, Y., Paasio, H., Li, C., Semjonova, G. & Xiao, Y. (2026). *Exploring Human-AI Collaboration in E-Textile Design: A Case Study on Flex Sensor Placement for Shoulder Motion Detection*. Proceedings of the ACM on Human-Computer Interaction. (Accepted).
  - *Role:* First author. Investigated LLM-assisted sensor placement optimization for e-textile motion capture.
- **5.** Zhang, Y., **Liu, Z.**, He, Y., Ploetz, T. & Xiao, Y. (2026). *Fine-grained Motion Retrieval via Joint-Angle Motion Images and Token-Patch Late Interaction*. arXiv preprint arXiv:2603.09930. (Under Review).
  - *Role:* Co-author. Contributed to the late interaction retrieval framework and experimental design.
- **6.** Zhang, Y., **Liu, Z.** & Xiao, Y. (2026). *LingoMotion: An Interpretable and Unambiguous Symbolic Representation for Human Motion*. In 2026 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops). (Accepted).
  - *Role:* Co-author. Contributed to the design of the symbolic representation framework and experimental validation.
- **7. Liu, Z.**, Chen, S., Tavallaey, S.S., Heintz, F. & Zea, E. (2025). *AcademicRAG: A Knowledge Graph-Enhanced Framework for Intelligent Academic Resource Discovery*. Big Data Analytics & Applications, p.5.
  - *Role:* First author. Led the development of the KG-enhanced RAG framework during Master's thesis at ABB.
- **8. Liu, Z.** (2022). *User Adoption Analysis of Social Media Applications Based on NLP and Technology Acceptance Model*. 2022 5th International Conference on Data Science and Information Technology (IEEE-DSIT).
  - *Role:* Sole author.

Other research outputs

- **Software/Tool:** AcademicRAG (GitHub) - A comprehensive framework integrating Knowledge Graphs with RAG for academic literature retrieval.

Awards and honours

- **2025: Kaggle Competitions Master**
  - Awarded the Master tier (Ranked top 0.5% of the global data science community) for consistent high-performance in international machine learning competitions.
  - *Granting body:* Kaggle (Google).
- **2023: Overseas Study Scholarship for Outstanding Undergraduates**
  - Awarded for academic excellence during Bachelor's studies.
  - *Granting body:* Jiangxi University of Finance and Economics.

Other expertise of relevance

- **Technical Skills:**

- *Core:* Python (PyTorch), LLM Fine-tuning, RAG Pipelines, Knowledge Graphs.
- *Tools:* Docker, Git, Linux, HuggingFace Ecosystem.